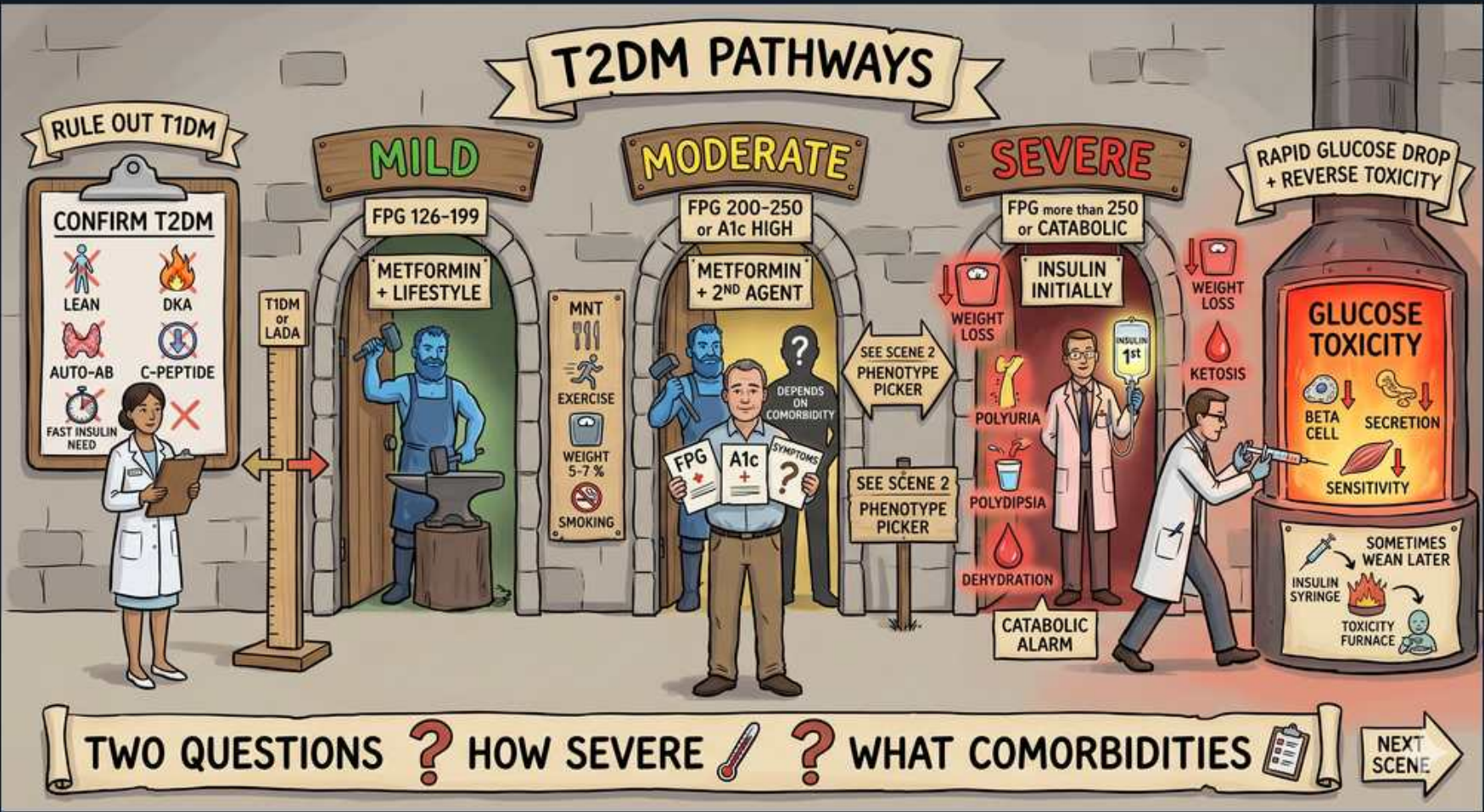


Diabetes — Scene 1: T2DM Severity Triage Gate



1. Two questions framework

WHAT YOU SEE

Bottom banner: 'Two questions — how severe? what comorbidities?'

WHAT IT ENCODES

Every new T2DM management decision answers two questions: (1) HOW SEVERE is the hyperglycemia (FPG/A1c level, catabolic features), which sets the INTENSITY of initial therapy, and (2) WHAT COMORBIDITIES exist (ASCVD, HF, CKD, obesity, hypoglycemia risk), which set the CHOICE of agent. Diagnostic cutoffs to anchor severity: A1c ≥6.5%, FPG ≥126, 2-hr OGTT ≥200, or random ≥200 with symptoms (confirm with a repeat unless unequivocally symptomatic). Mnemonic: 'severity sets the dose, comorbidity sets the drug.'

BOARD TRAP

Wrong answer: 'pick the drug class first and worry about how high the sugar is later.' Discriminator — if FPG >250 / A1c very high or catabolic features are present, you must START INSULIN regardless of which comorbidity-tailored agent you would otherwise choose. Severity gates the decision before phenotype does.

2. Rule out T1DM / LADA

WHAT YOU SEE

'Rule out T1DM': confirm T2DM — lean, DKA, autoantibodies, C-peptide, fast insulin need.

WHAT IT ENCODES

Before running the T2DM ladder, confirm it is actually T2DM. Red flags for autoimmune diabetes: LEAN body habitus, ketosis/DKA at presentation, rapid need for insulin, personal/family autoimmune disease. Confirmatory tests: positive islet autoantibodies (GAD65 most useful in adults, IA-2, ZnT8, IAA) and a LOW C-peptide (reflecting absent endogenous insulin). Mnemonic: 'lean + ketones + autoantibodies + low C-peptide = NOT type 2.'

BOARD TRAP

Wrong answer: 'a 35-year-old with a high A1c and recent weight loss has T2DM — start metformin.' Discriminator — lean adult with weight loss/ketosis and rapid insulin need is likely LADA or T1DM; check GAD65 antibodies and C-peptide. Metformin monotherapy is inadequate and dangerous if they are insulinopenic — they need insulin.

3. MILD pathway

WHAT YOU SEE

'MILD' arch: FPG 126-199 or A1c high; metformin + lifestyle.

WHAT IT ENCODES

MILD T2DM (FPG ~126–199, modest A1c elevation, asymptomatic) → start METFORMIN plus intensive lifestyle. Metformin reduces hepatic gluconeogenesis, is weight-neutral, causes no hypoglycemia, lowers A1c ~1–1.5%, and is dosed up to ~2000 mg/day (use XR for GI tolerance; check B12). Mnemonic: 'mild = metformin + move more.'

BOARD TRAP

Wrong answer: 'start insulin or dual oral therapy for a mildly elevated A1c.' Discriminator — for mild disease, monotherapy metformin + lifestyle is sufficient initial therapy; escalating to insulin/dual therapy is overtreatment and risks hypoglycemia/weight gain. Reserve intensification for moderate/severe or treatment failure.

4. MILD — MNT / lifestyle

WHAT YOU SEE

MNT, exercise, weight 5-7%, smoking cessation.

WHAT IT ENCODES

Lifestyle is foundational at every severity: medical nutrition therapy (MNT), ≥150 min/week moderate aerobic exercise + resistance training, 5–7% weight loss (this magnitude meaningfully improves glycemia and was the DPP target), and smoking cessation (smoking worsens insulin resistance and CV risk). Mnemonic: 'lose 5–7%, move 150, quit smoking.'

BOARD TRAP

Wrong answer: 'lifestyle change is optional once a drug is started.' Discriminator — MNT, exercise, weight loss, and smoking cessation continue alongside ALL pharmacotherapy and at every stage; they are never dropped. Drugs supplement, not replace, lifestyle.

5. MODERATE pathway

WHAT YOU SEE

'MODERATE' arch: FPG 200-250 or A1c high; metformin + 2nd agent.

WHAT IT ENCODES

MODERATE T2DM (FPG ~200–250, A1c roughly ≥1.5–2% above target, e.g. ≥9% but asymptomatic) → start DUAL therapy up front: metformin PLUS a second agent chosen by comorbidity phenotype. Combining at initiation reaches target faster than sequential single-agent steps. Mnemonic: 'A1c ≥1.5% over goal → start two drugs.'

BOARD TRAP

Wrong answer: 'always start metformin alone and wait 3 months even when A1c is 9.5%.' Discriminator — when baseline A1c is markedly above target (≥1.5–2% over goal) or FPG is in the 200s, initial combination therapy is preferred over metformin monotherapy. Sequential single-drug steps would leave the patient uncontrolled too long.

6. Phenotype picker handoff

WHAT YOU SEE

'See Scene 2 — Phenotype Picker'; FPG and A1c dials with question mark.

WHAT IT ENCODES

The identity of the second (and third) agent is decided by the comorbidity phenotype, detailed in the Phenotype Picker scene: ASCVD → GLP-1 RA or SGLT2i; HF → SGLT2i; CKD/albuminuria → SGLT2i ± GLP-1 RA; obesity → GLP-1 RA/tirzepatide; high hypo risk → avoid SU/insulin. Mnemonic: 'severity opened the door; phenotype picks who walks through.'

BOARD TRAP

Wrong answer: 'pick the second agent by which is cheapest or most potent on A1c.' Discriminator — second-agent selection is driven by the organ you most need to protect (CV/renal/weight/hypo profile), not by A1c potency or cost alone, whenever a compelling comorbidity is present.

7. SEVERE pathway — insulin

WHAT YOU SEE

'SEVERE' arch: FPG >250 or catabolic; insulin initially; weight loss, ketosis.

WHAT IT ENCODES

SEVERE T2DM (FPG >250–300, A1c >10%, OR catabolic features: weight loss, ketosis, marked polyuria/polydipsia) → start INSULIN initially to break glucose toxicity, often as basal insulin titrated to fasting glucose, with prandial added if needed. Once glucotoxicity resolves and beta-cell function recovers, many patients can be weaned to oral/non-insulin agents. Mnemonic: 'severe or symptomatic → insulin first.'

BOARD TRAP

Wrong answer: 'start metformin plus a GLP-1 RA in a patient with FPG 320, ketones, and 10-lb weight loss.' Discriminator — catabolic/ketotic or very high glucose signals profound insulinopenia (relative or possible T1/LADA); insulin is required first regardless of presumed type. Non-insulin agents act too slowly and won't cover the deficit.

8. Catabolic alarm

WHAT YOU SEE

Polyuria, polydipsia, dehydration, weight loss, ketosis → 'catabolic alarm.'

WHAT IT ENCODES

Catabolic features — unintentional weight loss, ketosis, polyuria, polydipsia, dehydration — indicate that insulin is insufficient to suppress lipolysis/proteolysis, i.e. significant insulin deficiency. This phenotype mandates insulin and should also prompt reconsideration of type (check antibodies/C-peptide if lean/atypical). Mnemonic: 'wasting + ketones = the tank is empty, refill with insulin.'

BOARD TRAP

Wrong answer: 'these symptoms are just uncontrolled T2DM; titrate the sulfonylurea.' Discriminator — catabolism means the beta cells can't meet demand, so a secretagogue that squeezes failing beta cells is the wrong tool; give exogenous insulin. Also screen for DKA and for underlying T1DM/LADA.

9. Wean insulin later

WHAT YOU SEE

'Sometimes wean later': insulin syringe → toxicity furnace.

WHAT IT ENCODES

Insulin started to overcome glucose toxicity in severe T2DM is often TEMPORARY: once chronic hyperglycemia is corrected, suppressed beta-cell secretion and insulin sensitivity recover, and the patient can frequently transition to metformin ± non-insulin agents. Mnemonic: 'put out the glucotoxic fire with insulin, then step down.'

BOARD TRAP

Wrong answer: 'once a T2DM patient is on insulin, they require it for life.' Discriminator — unlike T1DM, many T2DM patients started on insulin for glucotoxicity can be weaned after the toxicity reverses. Assuming lifelong insulin is incorrect for type 2 (it is true for type 1).

10. Glucose toxicity

WHAT YOU SEE

'Glucose toxicity' jar: beta cell, secretion, sensitivity all impaired.

WHAT IT ENCODES

Glucotoxicity = chronic severe hyperglycemia that further impairs BOTH beta-cell insulin SECRETION and peripheral insulin SENSITIVITY, creating a vicious cycle. Crucially it is REVERSIBLE: aggressive glucose lowering (usually insulin) restores secretion and sensitivity. Mnemonic: 'high sugar poisons the very cells that fight it — but the poison washes out.'

BOARD TRAP

Wrong answer: 'the beta-cell impairment in a newly diagnosed hyperglycemic patient is permanent.' Discriminator — glucotoxicity-driven beta-cell dysfunction is reversible with prompt glucose control, which is exactly why temporary insulin can later be withdrawn. Don't mistake reversible glucotoxicity for irreversible beta-cell loss.

11. Rapid glucose drop / reverse toxicity

WHAT YOU SEE

'Rapid glucose drop + reverse toxicity.'

WHAT IT ENCODES

Rapidly lowering glucose with insulin reverses glucotoxicity, restoring endogenous insulin secretion and sensitivity, which is the rationale for short-term insulin in severe presentation. (Caveat outside this scene: in long-standing uncontrolled retinopathy, very rapid lowering can transiently worsen retinopathy — lower steadily.) Mnemonic: 'break the toxicity fast, but respect the eyes.'

BOARD TRAP

Wrong answer: 'avoid lowering glucose quickly because it has no benefit.' Discriminator — prompt control DOES reverse glucotoxicity and is beneficial; the only nuance is gradual lowering when severe proliferative retinopathy is present. The general principle is to reverse glucotoxicity actively.

12. T1DM vs LADA door

WHAT YOU SEE

Side door labeled T1DM / LADA off the confirm-T2DM room.

WHAT IT ENCODES

If autoantibodies (GAD65 etc.) are positive with a low C-peptide, the patient exits the T2DM ladder into T1DM/LADA management: insulin-based therapy (LADA progresses to insulin dependence over months–years, faster than T2DM). Metformin/SGLT2i can be adjunctive in some, but autoimmune diabetes is fundamentally insulin-deficient. Mnemonic: 'antibody-positive + low C-peptide → take the insulin door.'

BOARD TRAP

Wrong answer: 'manage an antibody-positive lean adult exactly like T2DM with escalating orals.' Discriminator — LADA is autoimmune; oral secretagogues exhaust the failing beta cells and patients become insulin-dependent. Route them to insulin-based therapy, not the standard T2DM oral ladder.